LECTURE 1

Ex Evaluate $\lim_{x \to \infty} \frac{x^2 + x}{2x^2 + 5}$

$$= \lim_{x \to \infty} \frac{x^2/x^2 + x/x^2}{2x^2/x^2 + 5/x^2}$$

$$= \lim_{x \to \infty} \frac{1 + 1/x}{2 + 5/x^2}$$

$$= \frac{1 + 0}{2 + 0}$$

$$= \frac{1}{2}$$

Ex Evaluate $\lim_{x \to \infty} \frac{e^{2x} + e^x}{4e^{3x} + e^x}$

$$= \lim_{x \to \infty} \frac{e^{2x}/e^{3x} + e^x/e^{3x}}{4e^{3x}/e^{3x} + e^x/e^{3x}}$$

$$= \lim_{x \to \infty} \frac{1/e^x + 1/e^{2x}}{4 + 1/e^{2x}}$$

$$= \frac{0 + 0}{4 + 0}$$

$$= 0$$

Ex Evaluate $\lim_{x \to \infty} \frac{\ln(x)}{x}$ (type $\frac{\infty}{\infty}$)

$$\overset{LHR}{=} \lim_{x \to \infty} \frac{1/x}{1}$$

$$= \lim_{x \to \infty} \frac{1}{x}$$

$$= 0$$

(fundamental log limit)

limits as $x \to \infty$, hopital's rule

Ex

Evaluate $\int x e^{x^2} \, dx$

Let $u = x^2$. We have $du = 2x \, dx$

$\frac{1}{2} du = x \, dx$

$$= \int e^{x^2} x \, dx$$

$$= \int e^u \cdot \frac{1}{2} \, du$$

$$= \frac{1}{2} e^u + c$$

$$= \frac{1}{2} e^{x^2} + c$$

Ex

Evaluate $\int_1^2 \frac{(\ln(x))^2}{x} \, dx$

Let $u = \ln(x)$

$du = \frac{1}{x} \, dx$

When $x = 1$, $u = \ln(1) = 0$

$x = 2$, $u = \ln(2)$

$$= \int_1^2 (\ln(x))^2 \cdot \frac{1}{x} \, dx$$

$$= \int_0^{\ln(2)} u^2 \, du$$

$$= \frac{1}{3} (\ln(2))^3 - \frac{1}{3} (0)^3$$